

❖ **Project Title –**

Influencer Fraud Detection Using Anomaly Detection

❖ **Project Objective**

The goal of this project is to build an automated system that detects fraudulent influencer accounts using engagement metrics and anomaly detection models.

❖ **Dataset Description**

Dataset.

- 5000 influencer profiles
- Metrics collected:
 - username
 - followers count
 - following count
 - total posts
 - average likes
 - Average comment
 - Engagement rate
 - Growth rate

❖ Models Used

Explain models briefly:

- Isolation Forest
- Local Outlier Factor (LOF)

These unsupervised anomaly detection models identify statistical outliers in influencer engagement behavior.

❖ Fraud Detection Thresholds

Fraud Score	Risk Level	Decision
0–39	Genuine	Approve
40–79	Suspicious	Manual Review
80–100	High Risk	Reject

Thresholds help marketing teams automatically filter suspicious influencer accounts.

❖ Business Decisions

- Approve genuine influencers

- Flag suspicious accounts for manual review
- Reject high-risk influencers

Benefits:

- Avoid fake engagement
- Reduce marketing fraud
- Improve campaign ROI

❖ Model Evaluation

- Precision
- Recall
- F1-score
- Confusion Matrix

Limitation:

Since anomaly detection models detect statistical outliers rather than fraud labels, detected anomalies may not perfectly match manually labeled fraud profiles.

❖ Conclusion

The anomaly detection models identify statistical outliers rather than learning from fraud labels. Since the evaluation dataset uses heuristic fraud rules and

simulated influencer data, the detected anomalies may not perfectly align with labelled fraud profiles. However, the model successfully demonstrates an automated fraud risk scoring pipeline for influencer vetting.